

NAMD-LMI Tutorial

Hefei-NAMD with light-matter interaction support

Linjie Chen

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University of Science and Technology of China

Light Matter Interaction

When an atom is in the presence of an external *classical electromagnetic fields*, $\mathbf{p} \rightarrow \mathbf{p} - e\mathbf{A}$, Hamiltonian becomes:

$$\begin{aligned} H &= \frac{(\mathbf{p} - e\mathbf{A})^2}{2m_e} + V_0(\mathbf{r}) \\ &= \left(\frac{p^2}{2m_e} + V_0(\mathbf{r}) \right) + \left(-\frac{e\mathbf{A} \cdot \mathbf{p}}{m_e} + \frac{e^2 \mathbf{A}^2}{2m_e} \right) \\ &= H_0 + H_1(t) \end{aligned}$$

1. $\mathbf{A}$: vector potential, $\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}$;
2. $V_0(r)$: ground-state potential for atom;
3. H_0 : ground-state Hamiltonian;
4. H_1 : light-matter interaction (LMI) term;
5. $\mathbf{A}^2$ term omitted: $H_1 \approx \frac{e\mathbf{A} \cdot \mathbf{p}}{m_e}$
6. Electric dipole approximation: $\mathbf{A}(\mathbf{r}, t) = \mathbf{A}(t)$.

In NAMD, wavefunction of system expanded as

$$|\Psi\rangle = \sum_j |\phi_j\rangle \langle \phi_j | \Psi \rangle = \sum_j c_j |\phi_j\rangle$$

then the system is propagated via

$$i\hbar \frac{\partial |\Psi(\mathbf{r}, \mathbf{R}, t)\rangle}{\partial t} = H^{\text{tot}}(\mathbf{r}, \mathbf{R}, t) |\Psi(\mathbf{r}, \mathbf{R}, t)\rangle$$

where the total Hamiltonian is given by

$$H^{\text{tot}}(\mathbf{r}, \mathbf{R}, t) = H^0(\mathbf{r}, \mathbf{R}, t) + H^{\text{LMI}}(\mathbf{r}, \mathbf{R}, t)$$

the propagation equation for the expansion coefficients

$$\begin{aligned} \frac{\partial c_{j(t)}}{\partial t} &= \sum_k \left[i\hbar^{-1} \langle \phi_j | H^{\text{tot}} | \phi_k \rangle + \left\langle \phi_j \left| \frac{\partial}{\partial t} \right| \phi_k \right\rangle \right] c_{k(t)} \\ &= \sum_k \left[i\hbar^{-1} H_{jk}^0 \delta_{jk} + i\hbar^{-1} H_{jk}^{\text{LMI}} + T_{jk} \right] \\ &\quad H_{jk}^{\text{LMI}} = -\frac{e}{m_e} \langle \phi_j | \mathbf{p} | \phi_k \rangle \cdot \mathbf{A}(t) \quad T_{jk} = \langle \phi_j | \frac{\partial}{\partial t} | \phi_k \rangle \end{aligned}$$

Basic workflow

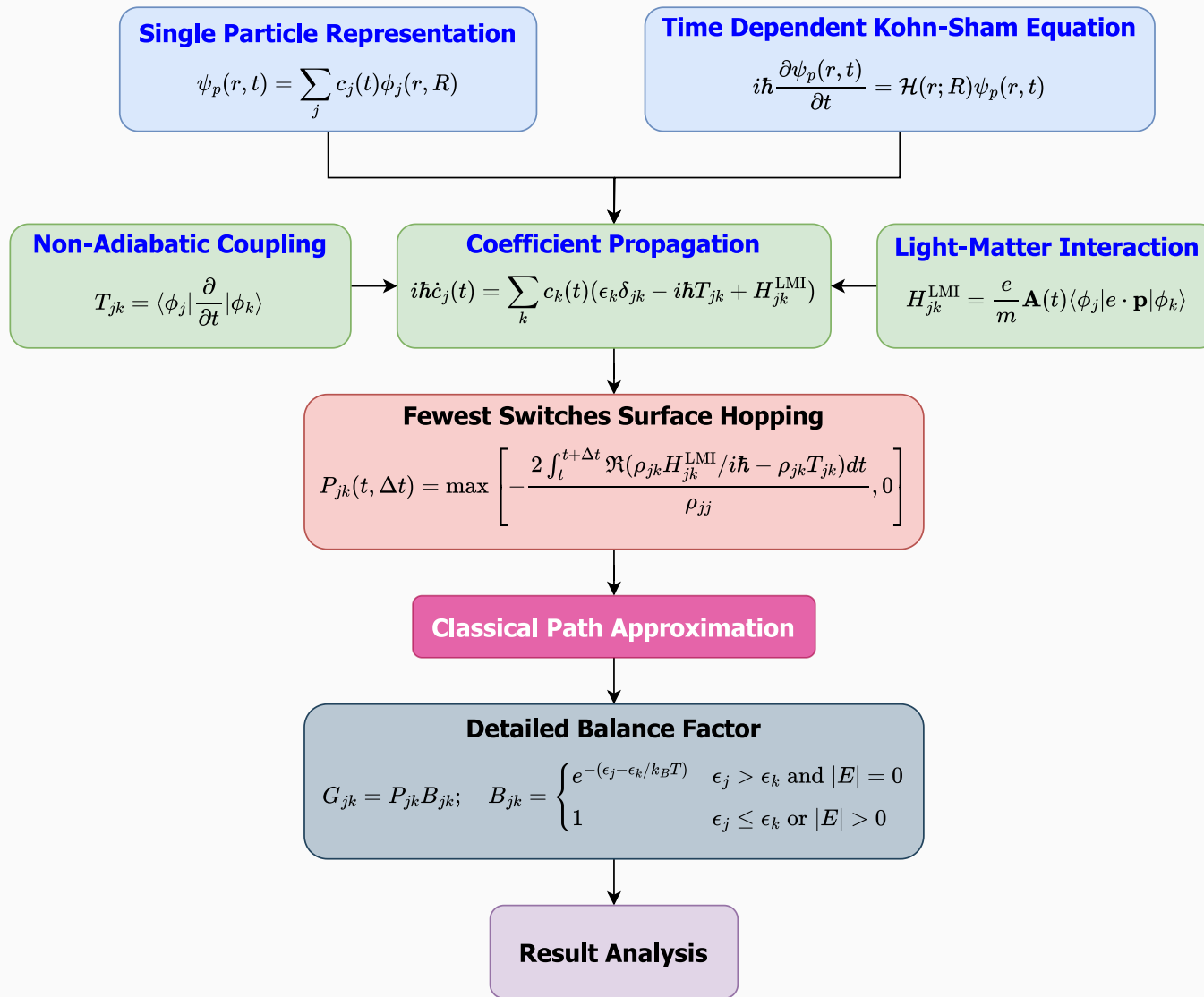


Figure 1: Basic workflow of NAMD-LMI' algorithm

Installation of NAMD-LMI

1. Download from GitHub Release (**Recommended**)

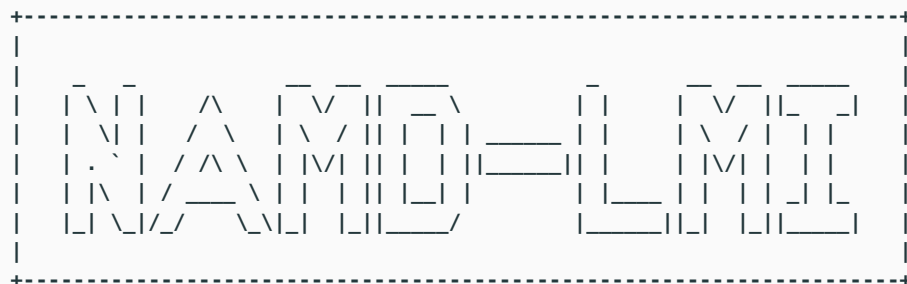
<https://github.com/Ionizing/NAMD-LMI/releases>

- `namd_lmi-0.3.1-linux-x86_64-mkl-system.tar.gz` (**Recommended**)
- `namd_lmi-0.3.1-linux-x86_64-mkl-static.tar.gz`
- `namd_lmi-0.3.1-macos-x86_64-openblas-static.tar.gz`
- `namd_lmi-0.3.1-macos-aarch64-openblas-static.tar.gz`
- `namd_lmi-0.3.1-windows-x86_64-mkl-static.zip`

2. Build from scratch (If necessary)

- See <https://ionizing.github.io/NAMD-LMI/Installation.html>

```
$ namd_lmi
```



Welcome to use namd!

```
current version: 0.3.1
git hash:       d7475d6
author(s):      Ionizing
host:           aarch64-apple-darwin
built time:     2025-04-11 23:04:51 +08:00
```

Usage: `namd_lmi <COMMAND>`

Commands:

```
waveslice  Slicing WAVECARs and PROCARs from pre-calculated AIMD
trajectory
  nac       Calculate non-adiabatic coupling (NAC) including  $\langle j | d/dt | k \rangle$  and momentum matrix  $\langle i | p | j \rangle$ 
  hamil     Generate the Hamiltonian from NAC according to config file
  surfhop   Perform the surface-hopping process with given Hamiltonian
            file and config file
  help      Print this message or the help of the given subcommand(s)
```

Options:

```
-h, --help    Print help (see more with '--help')
-V, --version  Print version
```

Workflow-VASP: Make supercell

```
#!/usr/bin/env python3
from ase.io import read
from ase.build import make_supercell
M = [[4,2,0],
      [0,3,0],
      [0,0,1]]
primcell = read("POSCAR.prim")
supcell = make_supercell(primcell, P=M, order="atom-major")
supcell.write("POSCAR.sup", vasp5=True, direct=True)
```

Mo	S		
1.000			
3.320	0.000	0.000	
-1.660	2.875	0.000	
0.000	0.000	15.000	

Mo	Se		
1	2		

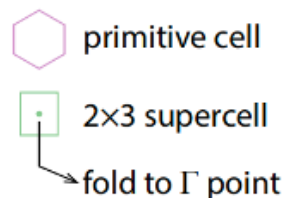
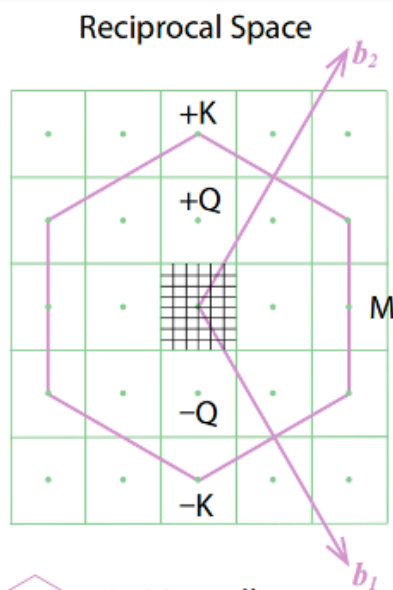
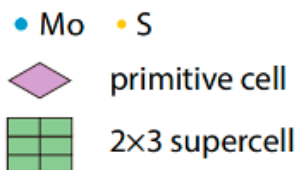
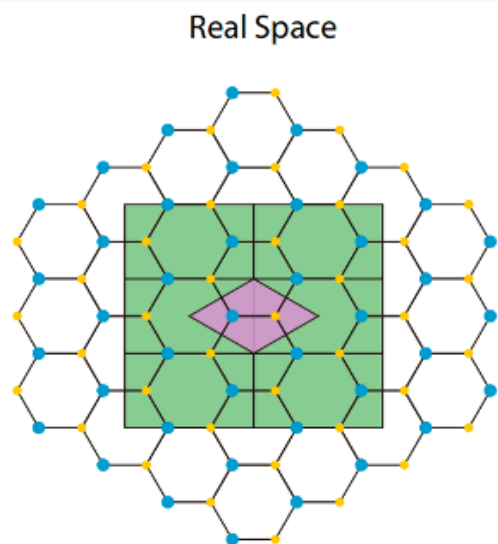
Direct			
0.6564	0.3282	0.5005	
0.9897	0.9948	0.3892	
0.9897	0.9949	0.6117	



Mo	Se		
1.000			
9.9600	5.7504	0.0000	
-4.9800	8.6256	0.0000	
0.0000	0.0000	15.0000	

Mo	Se		
12	24		

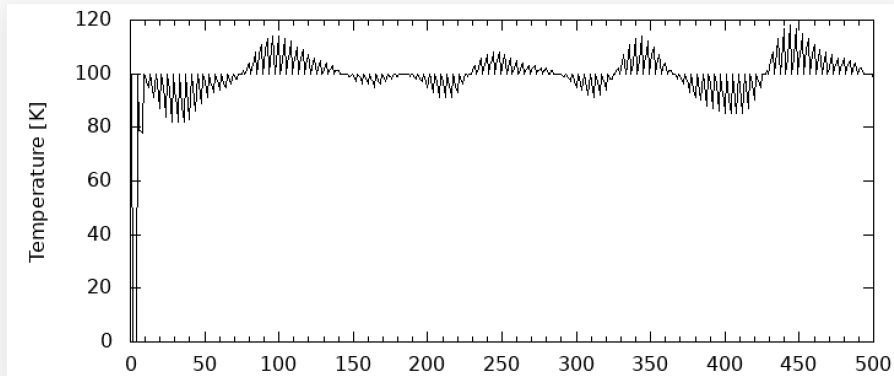
Direct			
0.1641	0.0000	0.5005	
0.1641	0.3333	0.5005	
0.1641	0.6667	0.5005	
0.4141	0.1667	0.5005	
0.4141	0.5000	0.5005	
.....			



Workflow-VASP: NVT & NVE

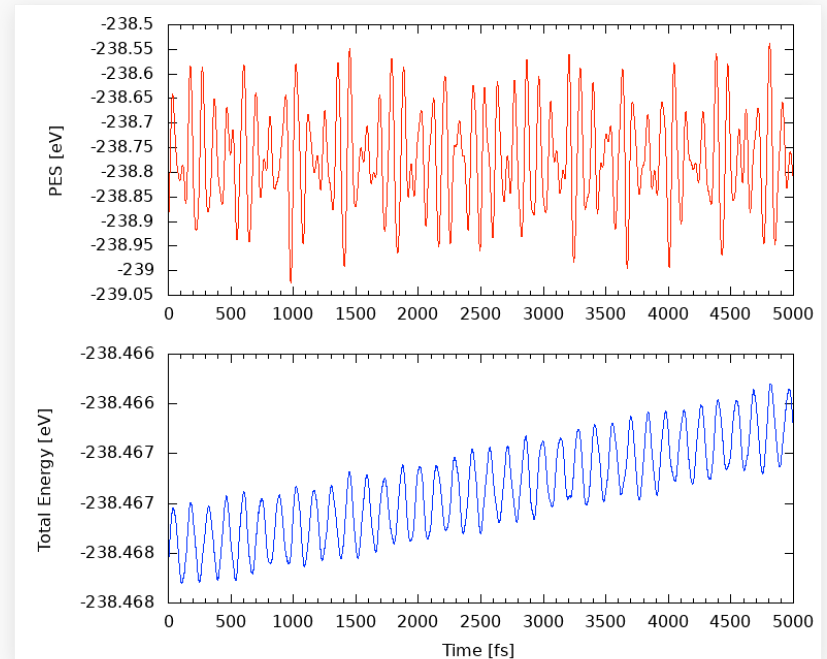
- NVT

```
...  
ISYM   = 0    # turn off symmetry  
IBRION = 0    # turn on MD  
NSW    = 500  # No. of ionic steps  
POTIM  = 1    # time step 1.0 fs  
SMASS  = -1   # Canonical  
NBLOCK = 4    # Rescale every 4 steps  
TEBEG  = 100  # Begin temperature  
TEEND  = 100  # End temperature  
...
```



- NVE

```
...  
ISYM   = 0    # turn off symmetry  
IBRION = 0    # turn on MD  
NSW    = 5000 # No. of ionic steps  
POTIM  = 1    # time step 1.0 fs  
SMASS  = -3   # Microcanonical  
NBLOCK = 1    # XDATCAR contains every step  
...
```



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Detailed workflow

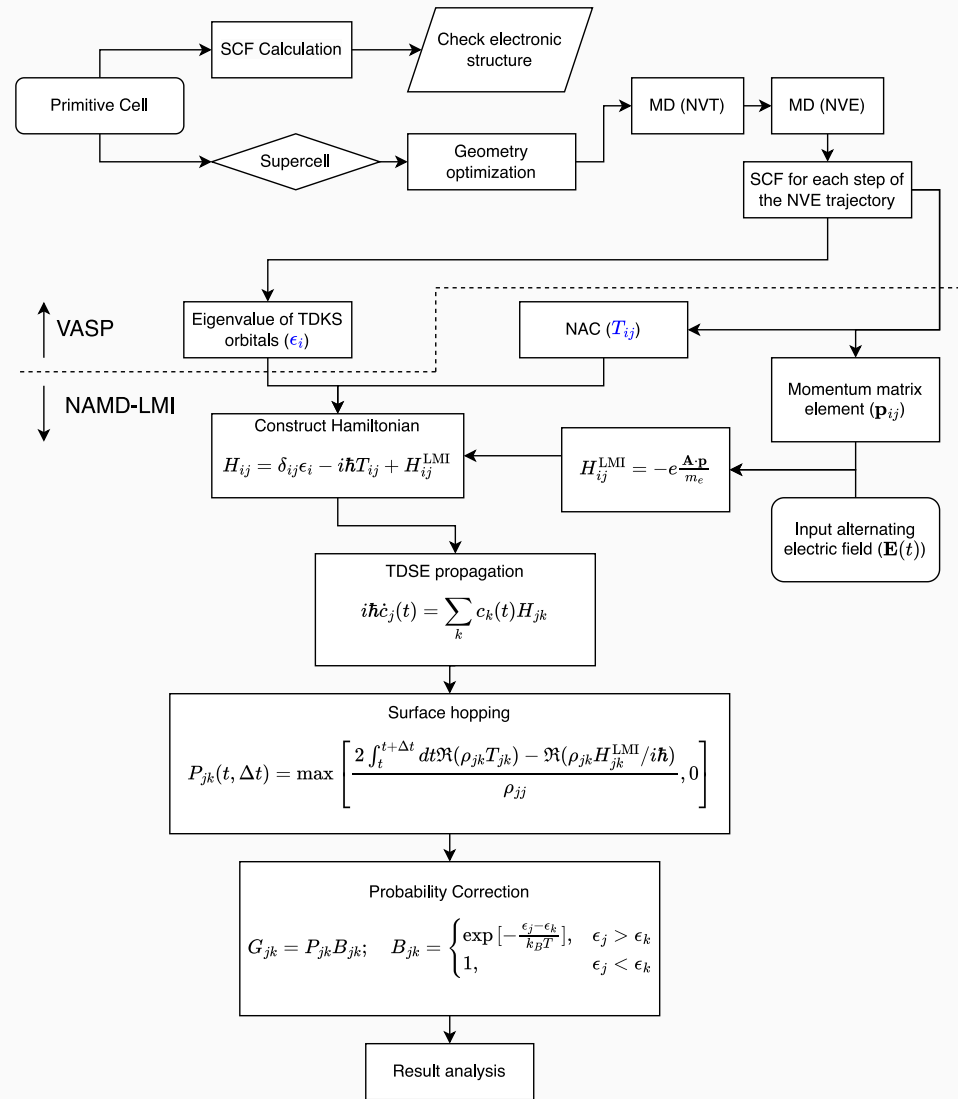


Figure 4: Detailed workflow of NAMD-LMI